



3D interpenetrating assembly of partially oxidized MXene confined Mn–Fe bimetallic oxide for superior energy storage in ionic liquid

Minjie Shi ^{a,1}, Mugilan Narayanasamy ^{a,c,1}, Cheng Yang ^b, Liping Zhao ^b, Jintian Jiang ^a, Subramania Angaiah ^c, Chao Yan ^{a,*}

^a School of Materials Science and Engineering, Jiangsu University of Science and Technology, Zhenjiang, 212003, PR China

^b State Key Laboratory for Metallic Matrix Composite Materials, School of Materials Science and Engineering, Shanghai Jiao Tong University, Shanghai, 200240, PR China

^c Electro-Materials Research Laboratory, Centre for Nanoscience and Technology, Pondicherry University, Puducherry, 605014, India

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ABSTRACT

Owing to unique diatomic synergistic effect and electron-occupied states at Fermi level, Mn–Fe bimetallic oxides hold great promise for high-efficiency supercapacitors (SCs). However, their energy storage behaviors are exploited in the aqueous electrolytes with narrow potential window, consequently leading to the low energy density of SCs. Herein we put forward a novel electrode (Ti₃C₂T_x@MFNDs) with a 3D interpenetrating assembly by confining MnFe₂O₄ nanodots (MFNDs) in hierarchically layered Ti-MXene (Ti₃C₂T_x), which shows superior energy storage in high-voltage ionic liquid (IL) electrolyte. Remarkably, 3D architecture is effectively constructed by the confinement of MFNDs in partially oxidized Ti₃C₂T_x with high conductivity and abundant active sites, providing multiple and continuous conductive paths for the efficient charge transport, as well as improving the structural stability of MFNDs in the electrode, demonstrated by ex-situ XRD analysis. Furthermore, the Ti₃C₂T_x@MFNDs electrode exhibits a high diffusion coefficient ($1.04 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$) and good wettability in IL electrolyte, indicating its superior IL ion dynamics. As a proof of concept, flexible ionogel SCs (FISCs) are fabricated, presenting high energy density (62.95 Wh kg⁻¹), high power density, remarkable rate capability and long-term durability. Such FISCs can be also charged by harvesting sustainable energy and effectively supply power for practical applications.

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1. Introduction

Benefiting from efficient electron-occupied states at the Fermi level and unique diatomic synergistic effect, bimetallic oxides with enhanced redox couples have received considerable attention in the field of energy storage applications [1–3]. Among them, as a novel spinel ferrite, Mn–Fe bimetallic oxide (MnFe₂O₄) has been deemed to be one of the promising electrodes for high-efficiency supercapacitors (SCs), which is owing to its multiple valence, high electrochemical activity and fast valence-state response capability [4–6]. Recently, researches have been focused on the preparation of MnFe₂O₄ with various structures for SCs electrodes. It has been demonstrated that the resultant MnFe₂O₄ electrodes

dominantly showed the faster redox reaction and much higher pseudocapacitance than monometallic oxides (Mn_xO_y or Fe_xO_y) [7–9]. Although MnFe₂O₄ electrodes have been documented to be utilized in several high-performance SCs based on aqueous electrolytes (Na₂SO₄ KOH, KCl, etc.), it is far more challenging to attain high energy density of these devices due to the narrow electrochemical windows of aqueous electrolytes, which is limited by the low decomposition voltage of water around 1.23 V [10–12].

Ionic liquid (IL), a class of room-temperature molten salts, has been regarded as an attractive candidate to substitute aqueous electrolytes of SCs, which is owing to their unique properties, such as wide operation voltage (>2.5 V), nonflammability, low vapor pressure, and high thermal stability [13,14]. Currently, considerable efforts have been dedicated to the development of SCs using IL electrolytes, especially carbon-based electrical double-layer SCs (EDLCs) [15–17]. In contrast to EDLCs, pseudocapacitance resulting from reversible redox reactions of pseudocapacitive material-based electrodes can deliver much higher specific capacitance for SCs.

* Corresponding author.

E-mail address: chaoyan@just.edu.cn (C. Yan).

¹ Minjie Shi and Mugilan Narayanasamy contributed equally to this work.

However, although MnFe_2O_4 electrodes have been demonstrated to deliver large specific capacitance in aqueous electrolytes by virtue of redox reactions, so far the reports on MnFe_2O_4 -IL electrolyte system are relatively limited till date, which are mainly ascribed to the poor surface wettability and interfacial mass transport between MnFe_2O_4 electrodes and IL electrolyte with high viscosity. The sluggish ionic diffusion and charge transport extremely affect the electrochemical kinetics and energy storage behaviors of MnFe_2O_4 electrodes, thus hindering their application in the high-voltage IL electrolyte.

To overcome the obstacle, an effective approach is to confine the MnFe_2O_4 into a suitable conductive skeleton to improve its activity as well as the electrochemical properties. As a new family of transition metal carbides/nitrides, MXene not only exhibits a 3D hierarchically layered structure with ultrahigh conductivity [18–21], but also possesses numerous active sites on the surface of the individual conductive flakes [22–25]. These properties enable effective confinement of other electro-active materials in MXene to construct a 3D interpenetrating conductive assembly, which can provide multiple and continuous conductive pathways for the efficient transport of ions and electrons, as well as improve the structural stability of electrode involved in energy storage [26–30]. Very recently, Luo et al. designed a 3D interpenetrating conductive assembly by intertwining MnO_x and MXene stacks for the electrode material in high-performance energy storage devices [31]. Wang et al. prepared an efficient SCs electrode with three-dimensional interconnected porous network by confining NiCoAl-LDH nanosheets in MXene, delivering a high specific capacity of 627 C g^{-1} in 1 M KOH aqueous electrolyte [32]. Despite the progress mentioned above, the research of electrochemical storage behaviors and surface wettability of 3D assembly based on MXene confined electro-active materials (especially Mn–Fe bimetallic oxides) is still limited in the IL electrolyte.

In this work, a novel electrode ($\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$) with a 3D interpenetrating assembly by confining MnFe_2O_4 nanodots (MFNDs) in hierarchically layered Ti-MXene ($\text{Ti}_3\text{C}_2\text{T}_x$) has been developed, which shows superior energy storage in high-voltage IL electrolyte. On the one hand, partially oxidized $\text{Ti}_3\text{C}_2\text{T}_x$ possesses abundant functional groups on the surface of individual conductive flakes, acting as numerous active sites favorable for the absorption of IL ions, which greatly enhance the wettability of $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ electrode in the IL electrolyte. On the other hand, 3D interpenetrating architecture can be effectively constructed by the confinement of MFNDs in partially oxidized $\text{Ti}_3\text{C}_2\text{T}_x$ with high conductivity, providing multiple and continuous conductive paths for the efficient charge transport. Furthermore, ex-situ XRD analysis and cycling test result both demonstrate the durable structural stability of $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ electrode during the charging-discharging process in the IL electrolyte. As a proof of concept, a kind of high-performance flexible ionogel SCs (FISCs) have been fabricated, presenting a high energy density (62.95 Wh kg^{-1}), a high power density (7000 W kg^{-1}), as well as remarkable rate and cycling performance. For a real application, the FISCs can be charged by harvesting sustainable energy and effectively power to various electronics.

2. Experimental

2.1. Synthesis of $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$

All of the reagents were analytical grade without further purification. Firstly, hierarchically layered Ti-MXene ($\text{Ti}_3\text{C}_2\text{T}_x$) was derived from MAX phase (Ti_3AlC_2) via a chemical etching technique as follow: Ti_3AlC_2 powder (400 mesh, 98.8%) was purchased from 11 technology Co., Ltd, PR China. 0.5 g of Ti_3AlC_2 was slowly added

into 15 mL of 40 wt% hydrofluoric acid (HF) solution at 40°C under magnetic stirring for 24 h to completely exfoliate the Al layers. Subsequently, the obtained $\text{Ti}_3\text{C}_2\text{T}_x$ was washed repeatedly with deionized water till the pH of the supernatant reached 6–7, separated by centrifugation and dried in vacuum at 60°C for 12 h. After that, 80 mg $\text{Ti}_3\text{C}_2\text{T}_x$ was uniformly dispersed in 80 mL of ethanol under sonication for 1 h. Then, 1 mM $\text{Mn}(\text{NO}_3)_2$ and 2 mM $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ were dissolved separately in the above solution and stirred for 10 min. 30 wt% ammonia aqueous solution was used to adjust the solution to pH 9–10. Subsequently, the mixed suspension was transferred to a 100 mL Teflon-lined stainless-steel autoclave and heated at 180°C for 16 h. Finally, the solid product was centrifuged, washed with deionized water, freeze-dried and then calcined at 300°C in N_2 atmosphere for 2 h to form $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$. For comparison, pure MFNDs was also prepared through a procedure similar to that mentioned above without addition of $\text{Ti}_3\text{C}_2\text{T}_x$.

2.2. Characterization methods

Field emission scanning electron microscope (FESEM, FEI Sirion 200) and Transmission electron microscopy (TEM, JED-2300F) were used to characterize the morphology of the samples. X-ray diffraction (XRD) patterns were characterized on a powder XRD system with a Cu K α radiation. X-ray photoelectron spectroscopic (XPS) measurements were performed on a Kratos AXIS Ultra DLD spectrometer with an Al K α X-ray source. The surface tension of 1-ethyl-3-methylimidazolium tetrafluoroborate (EMIMBF $_4$) IL was tested using a surface tensiometer (K100, Germany). The wettability behaviors of the MFNDs and $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ electrodes in the IL were measured using a contact angle measuring instrument (DCA300, Germany), in which the surface free energy was determined using conventional two-liquid-phase method.

2.3. Electrochemical measurements

All the electrochemical measurements were carried out using a VMP3 multi-functional electrochemical analysis equipment (Bio-Logic, France). Electrochemical behaviors were investigated by the mean of the cyclic voltammetry (CV), galvanostatic charge-discharge (GCD) and electrochemical impedance (EIS) methods. The CV and GCD tests were measured at different scan rates and current densities. The EIS plots were performed in the frequency ranging from 0.01 Hz to 100 kHz with 5 mV AC amplitude. For the three-electrode systematic method of single electrode in EMIMBF $_4$ IL electrolyte, the platinum (Pt) foil and Ag/AgCl electrodes were used as the counter and reference electrodes, respectively. The working electrode was prepared by mixing the resultant samples (MFNDs or $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$) with acetylene black and polyvinylidene fluoride (PVDF) as the binder dissolved in N-methyl-2-pyrrolidone (NMP) in a weight ratio of 8:1:1. The slurry was pasted on nickel foam substrate using a medical blade, which was then dried in a vacuum oven at 90°C overnight to obtain electrode. Each electrode contains active material with a mass loading of about 2 mg/cm^2 . For FISCs application, a two-electrode cell configuration was assembled using IL gel (ionogel) electrolyte, in which graphite paper functioned as flexible current collector, along with $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ and commercial activated carbon (AC) served as negative and positive electrodes, respectively. After that, the ionogel electrolyte was slowly sandwiched between two electrodes, which were pressed together under a pressure of $\sim 1 \text{ MPa}$ for 1 h. Ionogel electrolyte was prepared with 0.6 g Poly (vinylidene fluoride-hexafluoropropylene) (P(VDF-HFP)), 2 g EMIMBF $_4$ IL and 6 mL of acetone solution.

2.4. Calculations for the electrochemical tests

The specific capacitance of a single electrode (C_{sp}) was calculated from the anodic scan of the CV curve on the basis of Equation (1):

$$C_{sp} (Fg^{-1}) = \frac{\int i \cdot dt}{m \cdot V} \quad (1)$$

Here, i is the current changed by time t , m is the mass of a single electrode and V is the voltage window of the CV scan.

According to the Yury's previous reports [33,34], when the GCD curve is nonlinear, the capacity of a single electrode, instead of the capacitance, is evaluated using

$$C (mAh g^{-1}) = \frac{I \cdot \Delta t}{3.6} \quad (2)$$

where Δt (s) is the discharging time, I ($A g^{-1}$) is the constant mass current density.

The energy density of assembled FISCs was calculated on the basis of the discharge branch of GCD curve by using Equation (3) and the power density was further evaluated by using Equation (4).

$$E = \frac{1}{3.6} \cdot I_c \left(\int_{V_0}^{V_{max}} t dV \right) \quad (3)$$

$$P = 3600 \cdot E / \Delta t \quad (4)$$

where E ($Wh kg^{-1}$) is the energy density, P ($W kg^{-1}$) is the power density, I_c ($A g^{-1}$) is the constant discharge current density, Δt (s) is the discharge time, V_0 to V_{max} is the voltage window of the full FISCs.

3. Results and discussion

Ti-Mxene ($Ti_3C_2T_x$) has been prepared by selective etching the Al layers in Ti_3AlC_2 using hydrofluoric acid (HF). After etching, the stacked multilayer sheets of $Ti_3C_2T_x$ are successfully formed as shown in the FESEM image of $Ti_3C_2T_x$ (Fig. 1a), in which smooth surface and sharp edge are presented in the pristine accordion-like $Ti_3C_2T_x$, indicating that the HF treatment leads to the delamination of $Ti_3C_2T_x$ by spreading apart the basal planes with a structure similar to that of exfoliated graphite [31,35]. Owing to the $Ti_3C_2T_x$ with a hierarchically layered structure, a large amount of $MnFe_2O_4$ nanodots (MFNDs) are confined in the multilayer sheets of $Ti_3C_2T_x$ to form $Ti_3C_2T_x@MFNDs$ with 3D interpenetrating assembly. As shown in Fig. 1b, $Ti_3C_2T_x$ stacks become relatively rough and thick, resulting in undulated surface and blurred edge, indicating the growth of MFNDs on the surface of $Ti_3C_2T_x$. Energy-dispersive spectroscopy (EDS) mapping shows the elemental distribution of $Ti_3C_2T_x@MFNDs$ (Fig. 1c). Homogeneous distributions of Ti, C, Mn, and Fe are clearly observed, which confirms the successful decoration of the $Ti_3C_2T_x$ stacks with MFNDs. As seen from the TEM image of $Ti_3C_2T_x@MFNDs$ in Fig. 1d, numerous MFNDs with the average size of 10 nm are attach on the sheets of $Ti_3C_2T_x$, indicating the interspace between $Ti_3C_2T_x$ flakes is also covered by MFNDs. High-resolution TEM image in Fig. 1e shows the lattice fringes with spacing of 1 nm corresponding to the (002) plane of $Ti_3C_2T_x$ [36], whereas the lattice fringes with spacing of 0.49 nm belonging to the (111) plane of $MnFe_2O_4$ in MFNDs [37]. Furthermore, XRD pattern (Fig. 1f) exhibits noticeable (002) peak of $Ti_3C_2T_x$, as well as typical characteristic peaks indexed to iwakiite $MnFe_2O_4$ (JCPDS no. 38–0430) of MFNDs, which are consistent with the HRTEM result of $Ti_3C_2T_x@MFNDs$. Noteworthy, compared with the sharp and narrow peaks, there is an obvious broad peak located at around 20° in the pattern of $Ti_3C_2T_x@MFNDs$. This broad peak not only corresponds to the (111) characteristic peak of iwakiite $MnFe_2O_4$, but also results from the (002) plane of carbon. Which is mainly due to the amorphous carbon originating from the partial oxidation of

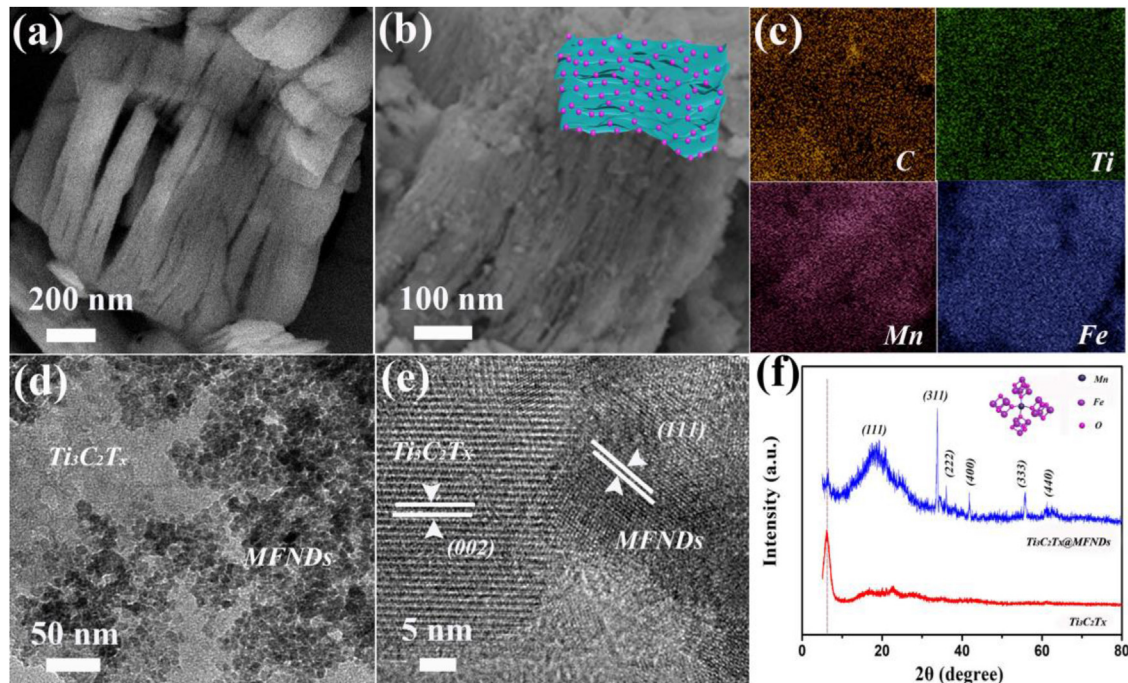


Fig. 1. (a) FESEM image of $Ti_3C_2T_x$. (b) FESEM image, (c) EDS mapping images, (d) TEM and (e) HRTEM images of $Ti_3C_2T_x@MFNDs$. Inset in b is the schematic diagram of $Ti_3C_2T_x@MFNDs$. (f) XRD pattern of $Ti_3C_2T_x$ and $Ti_3C_2T_x@MFNDs$. Inset is the molecular structure of MFNDs.

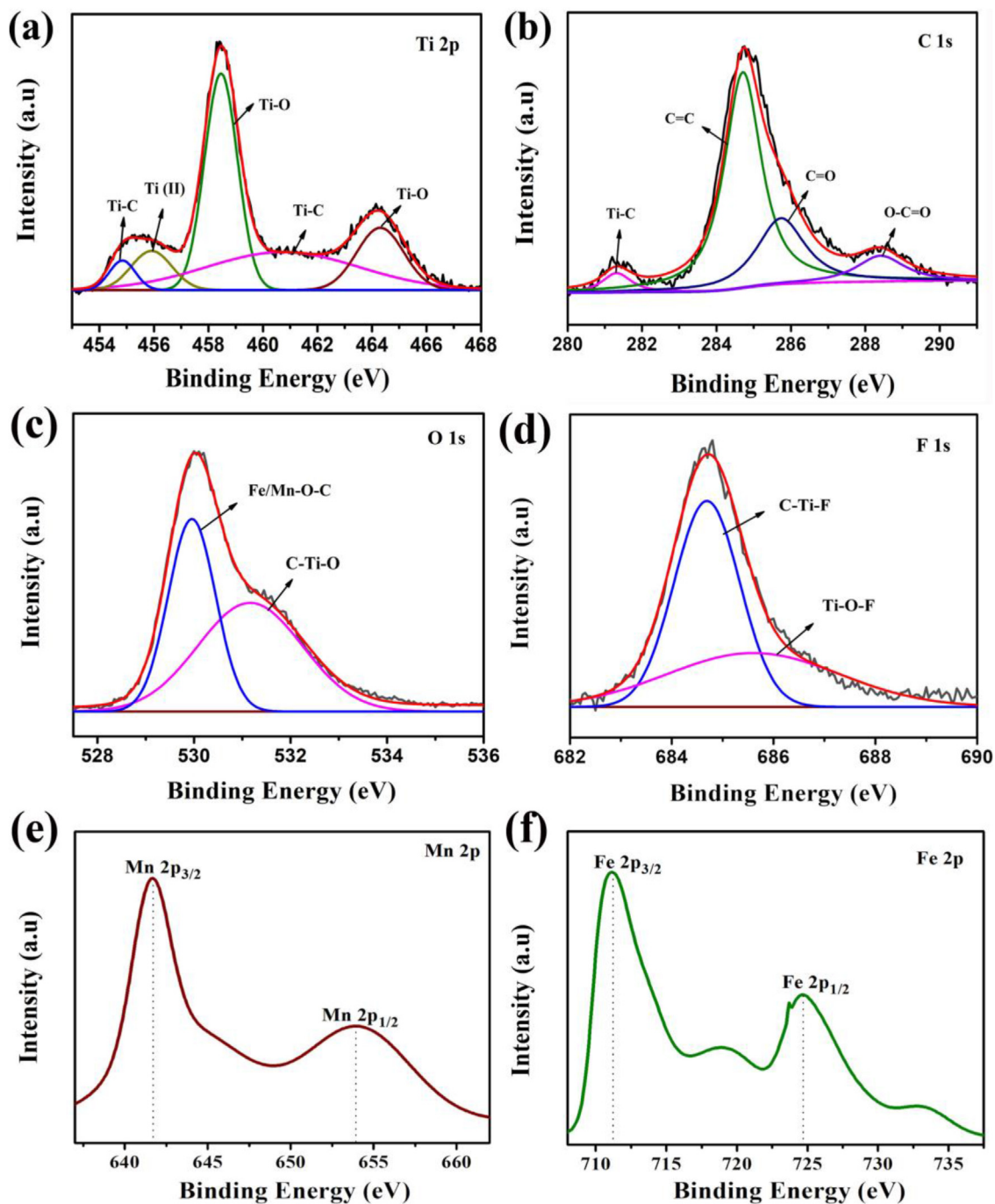


Fig. 2. XPS spectrum: (a) Ti 2p, (b) C 1s, (c) O 1s, (d) F 1s, (e) Mn 2p and (f) Fe 2p high-resolution spectra of $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$.

$\text{Ti}_3\text{C}_2\text{T}_x$ [38–40]. Besides, the Raman spectra of $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ are provided in Fig. S1.

XPS was acquired to determine the chemical composition of $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ composite. The high resolution Ti 2p spectrum can be deconvoluted into five peaks (Fig. 2a), corresponding to Ti–C (454.8 and 461.0 eV), Ti–O (458.6 and 464.2 eV) and Ti (II) (456.1 eV), which are similar to the previously reported literature [41,42]. As shown in Fig. 2b, the high resolution of C 1s spectrum can be divided into four peaks. The peaks with a bonding energy of 281.2 eV, 284.7 eV, 285.9 eV, and 288.5 eV correspond to the Ti–C, C=C, C=O and O–C=O, respectively [43,44]. This result indicates the partial oxidation of $\text{Ti}_3\text{C}_2\text{T}_x$ and the resultant formation of TiO_2

in the $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ during hydrothermal process. Although some TiO_2 exists in the $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$, it could provide additional capacitive storage in IL electrolyte. Meanwhile, remaining $\text{Ti}_3\text{C}_2\text{T}_x$ offers multiple and continuous conductive pathways for the efficient transport of ions and electrons. In Fig. 2c and d, the O 1s spectrum can be fitted into two subpeaks, including Fe(Mn)–O and C–Ti–O at 530.2 and 531.4 eV, respectively. Meanwhile, the F 1s spectrum can be also fitted into two subpeaks, including C–Ti–F and Ti–O–F at 684.9 and 685.8 eV, respectively. During the extraction of Al layer, $\text{Ti}_3\text{C}_2\text{T}_x$ could be spontaneously terminated with abundant functional groups, such as –O, –OH, and –F. As we known, suitable oxygen-containing functional groups not only

improve the surface wettability of electrode, but also provide more active sites to the rapid ionic kinetics of electrodes [45–47]. Besides, with respect to the Mn and Fe 2p spectra in Fig. 2e and f, there are not only two characteristic peaks at 642.0 and 653.3 eV related to the Mn 2p_{3/2} and Mn 2p_{1/2} of the Mn²⁺ oxidation state, but also two major peaks at 711.2 and 724.9 eV ascribed to the Fe 2p_{3/2} and Fe 2p_{1/2} of the Fe³⁺ oxidation state, which are characteristic of Mn²⁺ and Fe³⁺ of MnFe₂O₄ in MFNDs [7]. These results strongly indicate the successful formation of MFNDs confined in the Ti₃C₂T_x, where a large number of functional groups are existed in the surface of Ti₃C₂T_x, which can be beneficial for the adsorption and motion of electrolyte ions.

The electrochemical behaviors of pure MFNDs and Ti₃C₂T_x@MFNDs electrodes were evaluated using the three-electrode configuration in the EMIMBF₄ IL electrolyte. Fig. 3a shows the CV curves of MFNDs and Ti₃C₂T_x@MFNDs electrodes at a scanning rate of 50 mV s⁻¹. Prominent redox peaks are visible for the anodic and cathodic sweeps in the curves, indicating dominant pseudocapacitive storage of MFNDs and Ti₃C₂T_x@MFNDs electrodes in IL electrolyte. By contrast, the Ti₃C₂T_x@MFNDs electrode shows higher peak currents and larger CV area, thus revealing sufficient redox reaction of Ti₃C₂T_x@MFNDs electrode in IL electrolyte. More information about the electrochemical performance in IL electrolyte of pure Ti₃C₂T_x after hydrothermal treatment are provided in Fig. S2. EIS is a powerful technique that provided a wealth of information regarding the electrochemical kinetic characteristics of the electrode in IL electrolyte [48]. The point intersecting the real axis in the high frequency region displays the equivalent series resistance (*R*_s), which reveals the total inner resistance of the electrode. As shown in the inset of Fig. 3b, the *R*_s value of Ti₃C₂T_x@MFNDs electrode is lower than that of MFNDs electrode, owing to the high electrical conductivity of Ti₃C₂T_x in the Ti₃C₂T_x@MFNDs electrode. Moreover, the diameter of the semicircle in high frequency region is related to the charge-transfer resistance (*R*_{ct}), and the straight-line part in low frequency region shows Warburg impedance (*Z*_w), which is described as a diffusive impedance of the IL ions within the electrode. Clearly, the Ti₃C₂T_x@MFNDs electrode exhibits smaller semicircle and more vertical straight line (Fig. 3b), indicating the favorable kinetic

behaviors of ionic diffusion and charge transport of Ti₃C₂T_x@MFNDs electrode in IL electrolyte. For deeply analyzing the IL ion dynamics, the diffusion coefficient of IL ions in the electrode can be estimated using impedance data at a low frequency. In this frequency range, the magnitude of the impedance varies linearly with respect to $\omega^{-1/2}$, in accordance with Equation (5) [49,50]:

$$Z = \frac{L}{C_L(D \cdot \omega)^{1/2}} \quad (5)$$

where *D*, *C*_L, and *L* are the diffusion coefficient, low-frequency redox capacitance, and electrode thickness, respectively. The value of *C*_L can be determined from the plot of *Z*_{im} vs. 1/ω, which is linear with a slope of 1/*C*_L, in accordance with Equation (6):

$$Z_{im} = \frac{1}{C_L \omega} \quad (6)$$

According to Fig. S3, the *C*_L values of the Ti₃C₂T_x@MFNDs and MFNDs electrodes were calculated to be 17.73 and 52.74, respectively. Substituting these values as the slope in Equation (5), the diffusion coefficient of the IL ions in the Ti₃C₂T_x@MFNDs electrode is calculated to be 1.04 × 10⁻⁸ m² s⁻¹ (Fig. 3c), much higher than that in the MFNDs electrode (8.90 × 10⁻⁹ m² s⁻¹), further confirming the enhanced IL ion dynamics of the Ti₃C₂T_x@MFNDs electrode.

Wettability is one of the most crucial properties of the electrode in IL electrolyte [51,52]. For a droplet of liquid on a flat solid, the wetting behavior is determined by Young's equation [53,54]:

$$\gamma_{sl} = \gamma_{sg} - \gamma_g \cdot \cos \theta \quad (7)$$

According to Young's equation, there is a relationship among the contact angle (θ), the surface tension of the liquid (γ_{lg}), the surface free energy of the solid (γ_{sg}) and the interfacial tension (γ_{sl}) between the solid and liquid. Fig. 3d shows a schematic diagram of the components of a three-phase system which are relevant for the wetting process of the EMIMBF₄ IL drop in contact with the pure MFNDs and Ti₃C₂T_x@MFNDs electrodes. Obviously, the

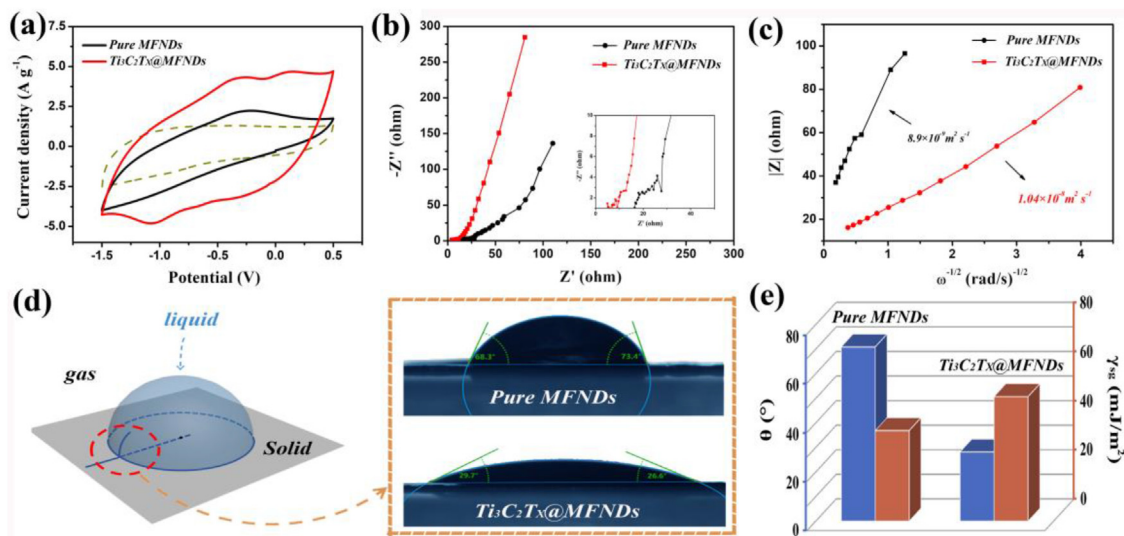


Fig. 3. (a) Comparison of CV curves scanning at 50 mV s⁻¹ of pure MFNDs and Ti₃C₂T_x@MFNDs electrodes in the EMIMBF₄ IL electrolyte. Yellow dashed line is the CV curve of Ti₃C₂T_x after hydrothermal method. (b) Nyquist plots and (c) diffusion coefficient curves of the pure MFNDs and Ti₃C₂T_x@MFNDs electrodes in the EMIMBF₄ IL electrolyte. (d) Contact angle of EMIMBF₄ IL droplets on the MFNDs and Ti₃C₂T_x@MFNDs electrodes. (e) The corresponding contact angle and solid surface free energy. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Ti₃C₂T_x@MFNDs electrode is almost completely wetted by the IL electrolyte, displaying significantly higher wettability than the MFNDs electrode. The outstanding wettability of the Ti₃C₂T_x@MFNDs electrode in IL electrolyte has been identified by its small θ (28.15°) and high γ_{sg} (50.68 mJ m⁻²) (Fig. 3e). Since the γ_{lg} of the EMIMBF₄ IL electrolyte is about 52.4 mJ m⁻², the γ_{sl} of the Ti₃C₂T_x@MFNDs electrode in the IL electrolyte is calculated to be 4.48 mJ m⁻², which is lower than a quarter of γ_{sl} for the MFNDs electrode (19.66 mJ m⁻²). The relatively lesser γ_{sl} value indicates the strong adhesion between the Ti₃C₂T_x@MFNDs electrode and IL electrolyte, which is beneficial from the abundant oxygenated surface functionality of Ti₃C₂T_x in the Ti₃C₂T_x@MFNDs electrode, thereby greatly improving the surface wettability and active sites of Ti₃C₂T_x@MFNDs electrode in IL electrolyte.

Fig. 4a shows all CV plots of Ti₃C₂T_x@MFNDs electrode in the EMIMBF₄ IL electrolyte at various scan rates with a comparatively wide electrochemical window of -1.5 V–0.5 V. By calculating from Equation (1), the specific capacitance of the Ti₃C₂T_x@MFNDs electrode based on total volume at 10 mV s⁻¹ can reach as high as 172.49 F g⁻¹ (Fig. 4b), which is comparable or superior to that previously reported electrodes in IL electrolytes [10,55–59]. To further analyze the electrochemical mechanism of Ti₃C₂T_x@MFNDs electrode in IL electrolyte in detail, the total charge storage was semi-quantitatively resolved by analyzing the current response of CV curves at different scan rates. As we known, the total capacitive behavior of Ti₃C₂T_x@MFNDs electrode comes from surface-reaction-controlled process and diffusion-controlled process.

Furthermore, the surface-reaction-controlled contribution to the total current is quantified by Equations (8) and (9) [22,60].

$$i = k_1 v + k_2 v^{1/2} \quad (8)$$

$$i/v^{1/2} = k_1 v^{1/2} + k_2 \quad (9)$$

Fig. 4c shows voltammetric response for the Ti₃C₂T_x@MFNDs electrode at scan rates of 50 mV s⁻¹, wherein the surface-reaction capacitance contribution to the total current is about 62.2%, which is distinguished by the blue shaded region. As shown in Fig. 4d, the contribution of the diffusion-limited process to the total capacity is noticeable at scan rates below 20 mV s⁻¹. However, at scan rates of ≥ 50 mV s⁻¹, the response comprises the surface capacitive effects rather than being diffusion-controlled, which is beneficial to achieving a favorable electrochemical behavior of the Ti₃C₂T_x@MFNDs electrode at high current densities [22].

Fig. 5a shows the GCD curve at various current densities of Ti₃C₂T_x@MFNDs electrode in the EMIMBF₄ IL electrolyte. The nonlinear curves without a plateau further show that Ti₃C₂T_x@MFNDs electrode mainly stores pseudocapacitive charge in IL electrolyte [33,34]. The charge stored in the Ti₃C₂T_x@MFNDs electrode, or the specific capacity, is evaluated by using GCD curves (on the basis of Equation (2)) at current densities from 0.5 A g⁻¹ to 4 A g⁻¹. As shown in Fig. 5b, the specific capacity of the Ti₃C₂T_x@MFNDs electrode is calculated to be 96.67 mAh g⁻¹ at 0.5 A g⁻¹ and 63.22 mAh g⁻¹ at 4 A g⁻¹, or a 65.39% capacity

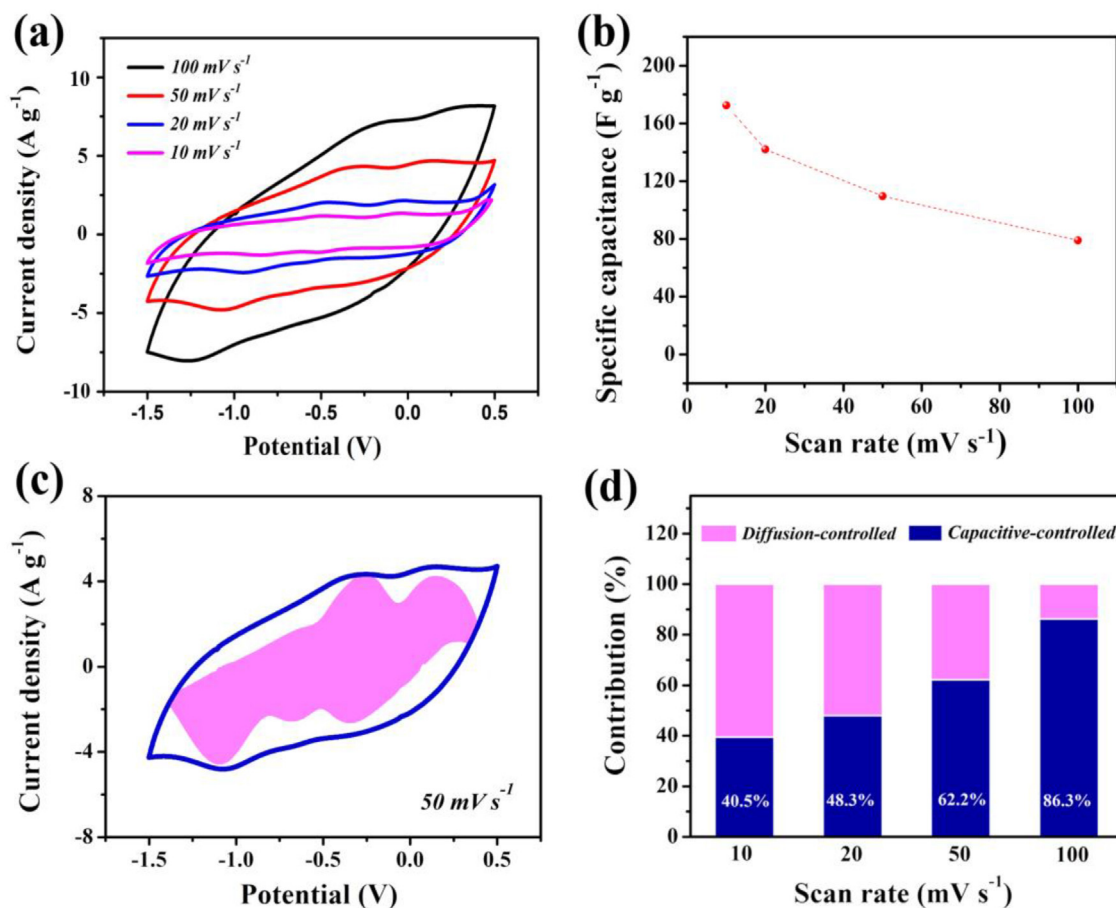


Fig. 4. (a) CV curves and (b) the corresponding specific capacitance at various scan rates of Ti₃C₂T_x@MFNDs electrode in IL electrolyte. (c) Voltammetric response at a scan rate of 50 mV s⁻¹ with the shaded region displaying the capacitive contribution. (d) The corresponding capacitive contributions at different scan rates.

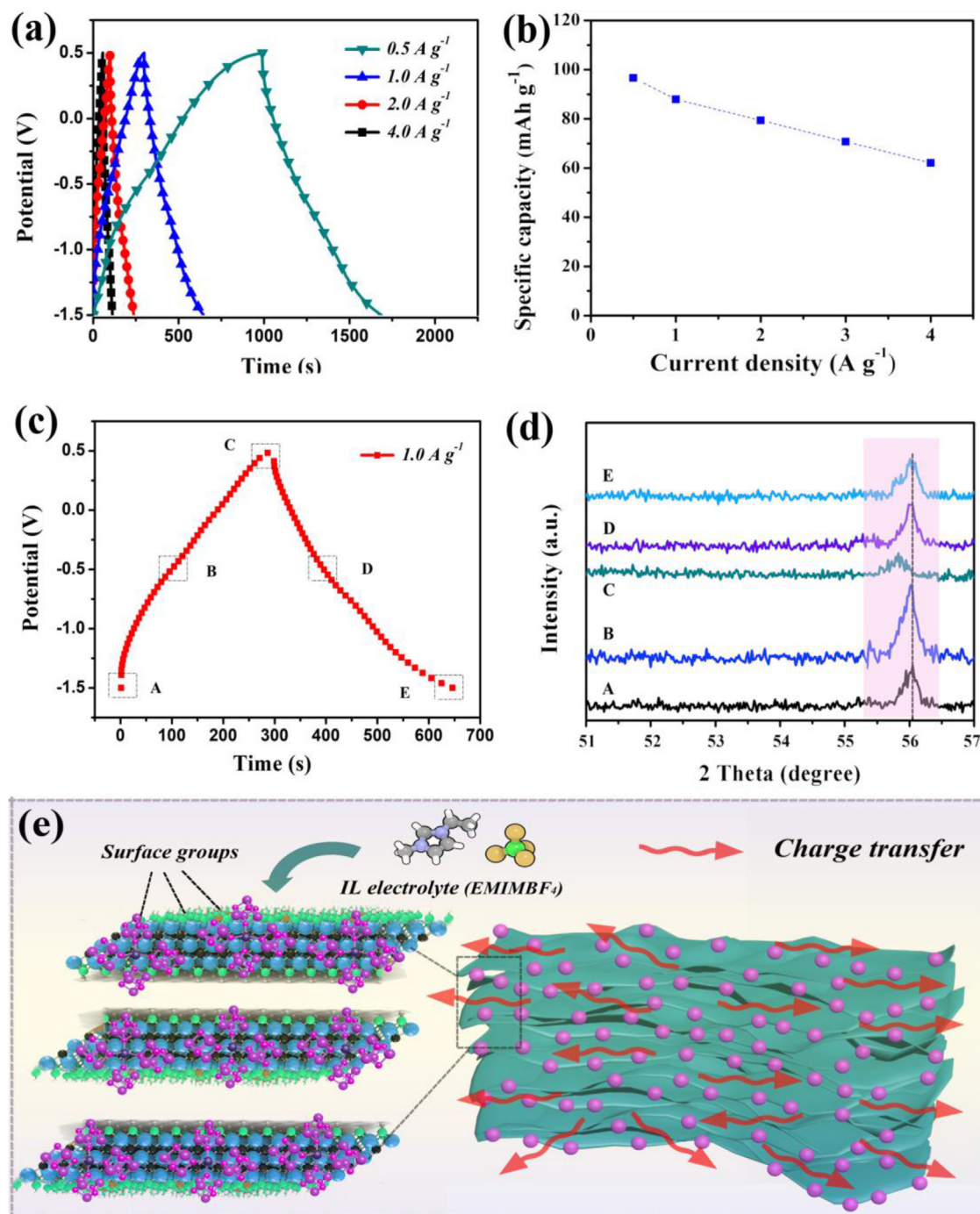


Fig. 5. (a) GCD curves and (b) specific capacity at various current density of $\text{Ti}_3\text{C}_2\text{Tx}@MFNDs$ electrode in IL electrolyte. (c) GCD curve at the current density of 1 A g^{-1} and (d) electrochemical ex-situ XRD study of $\text{Ti}_3\text{C}_2\text{Tx}@MFNDs$ electrode at various constant potentials in IL electrolyte. (e) Schematic diagram of ionic diffusion and charge transport in $\text{Ti}_3\text{C}_2\text{Tx}@MFNDs$ electrode during the charging-discharging process.

retention in IL electrolyte. As shown in Fig. 5c, the XRD diffraction patterns were recorded at various potentials (A, B, C, D and E) during the GCD cycle at a high current density of 1 A g^{-1} . In the case of charging section, the XRD patterns recorded in potential region A to C clearly reveal a shift of the reflections to lower 2θ values, suggesting an increase of the lattice parameters due to the expansion of the MnFe_2O_4 host lattice caused by the intercalation of IL ions into the electrodes [19,61]. When discharged from 0.5 V back to -1.5 V, a progressive shift of the MnFe_2O_4 (333) peak to higher 2θ values is occurred, indicating a decrease of the lattice spacing due

to the deintercalation of IL ions in the MnFe_2O_4 . In Fig. 5d, the (333) peak of MnFe_2O_4 for $\text{Ti}_3\text{C}_2\text{Tx}@MFNDs$ electrode initially observed at 56.05° at -1.5 V shifts to 55.82° when charged to 0.5 V. Afterwards, the (333) peak shifts back to 56.06° when discharged to -1.5 V, which is virtually the same as its original 2θ position, demonstrating a relatively reversible transition within $\text{Ti}_3\text{C}_2\text{Tx}@MFNDs$ electrode during the charge-discharge cycle [61,62]. Furthermore, the $\text{Ti}_3\text{C}_2\text{Tx}@MFNDs$ electrode shows good cycling stability with a high retention of 83.8% after 5000 cycles, which is better than that of (62.4%) MFNDs electrode in IL electrolyte (Fig. S4). Moreover,

FESEM image and XRD result of $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ electrode in IL electrolyte before and after cycling are supplied in Fig. S5. The ex-situ XRD analysis and cycling test result both indicate the durable structural stability of $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ electrode during the electrochemical process.

The superior electrochemical behaviors of the $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ electrode in IL electrolyte are schematically illustrated in Fig. 5e, which can be attributed to the following aspects. On the one hand, $\text{Ti}_3\text{C}_2\text{T}_x$ possesses abundant functional groups ($-\text{O}$, $-\text{OH}$, and $-\text{F}$) on the surface of individual conductive flakes, acting as numerous active sites favorable for the absorption of IL ions, which greatly enhance the surface wettability of $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ electrode in IL electrolyte. As a result, large amounts of IL ions could rapidly diffuse

into the interior of the $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ electrode (green arrow) to involve in the electrochemical reaction with MFNDs. On the other hand, 3D interpenetrating assembly of $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ electrode is effectively constructed by the ultrahigh conductive $\text{Ti}_3\text{C}_2\text{T}_x$ and MFNDs, which can provide multiple and continuous conductive paths for the efficient charge transport (red arrows). Furthermore, the confinement of MFNDs in $\text{Ti}_3\text{C}_2\text{T}_x$ can greatly inhibit the aggregation and dissolution of MFNDs during the cycling processes, rendering the high stability of the $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ electrode in IL electrolyte.

Similar to the IL electrolyte, ionogel as a solid-state electrolyte exhibits non-volatility, non-flammability, high thermal stability and wide electrochemical potential window [11,63]. For practical

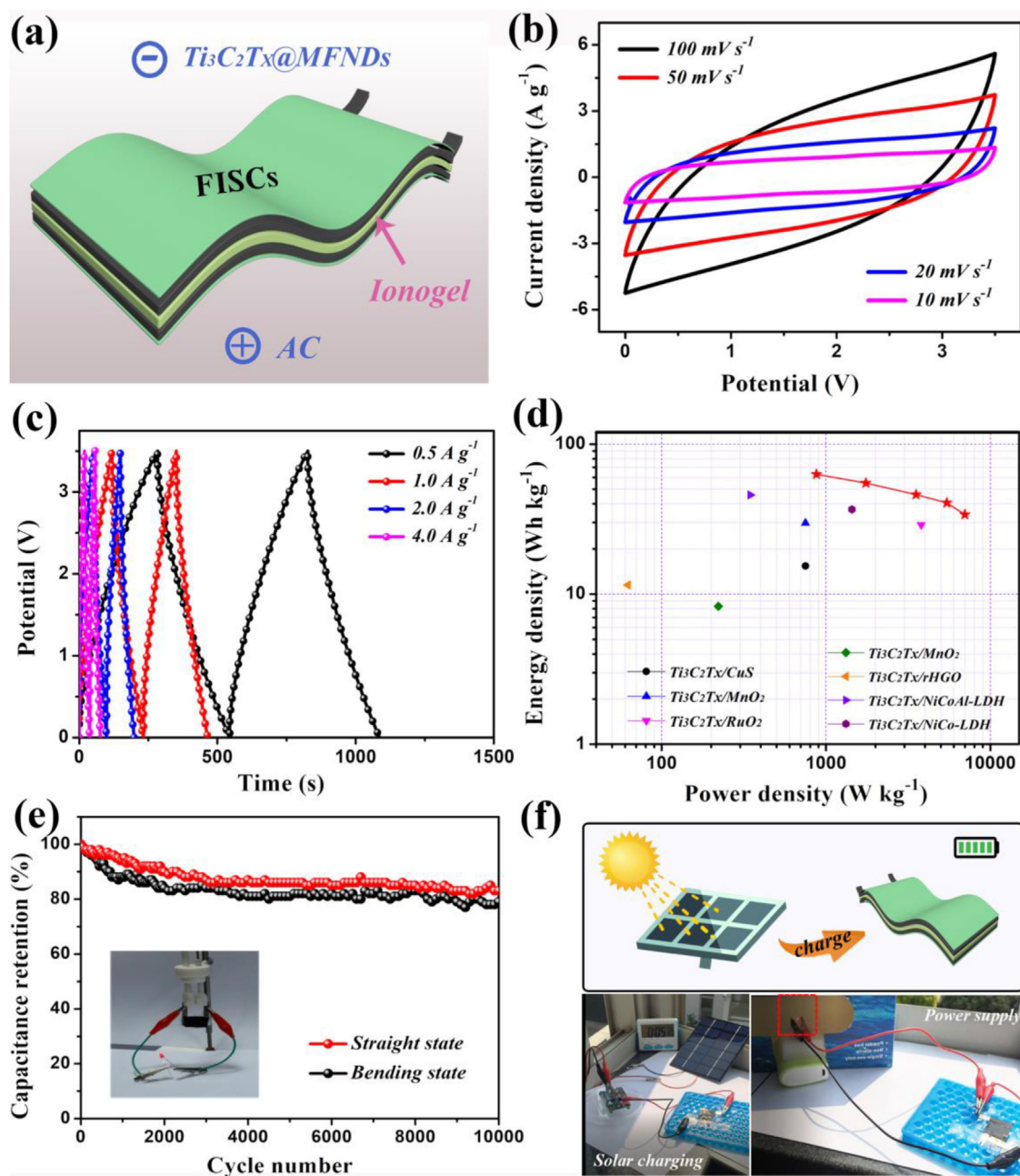


Fig. 6. (a) Fabrication procedure of $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ /AC FISCs. (b) CV curves, (c) GCD curves and (d) Ragone plot of the FISCs. (e) Cycle life of the devices in the straight and bending states at the current density of 2 A g^{-1} after 10 000 cycles. Inset shows the power supply of the FISCs. (f) Schematic diagram and real-time image of FISCs charged by harvesting sustainable solar energy in life.

application, a kind of flexible ionogel SCs (FISCs) have been fabricated, in which $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ and commercial activated carbon (AC, Figs. S6 and S7) pressed on flexible graphite paper were selected as negative and positive electrodes, respectively (Fig. 6a). The FISCs can be operated in a wide working voltage window between 0 and 3.5 V (Fig. 6b). Fig. 6c shows the GCD curves of the FISCs in the range of 0.5–4 A g⁻¹. By calculating from Equations (3) and (4), the FISCs deliver a maximum energy density up to 62.95 Wh kg⁻¹ at a power density of 875 kW kg⁻¹. Meanwhile, the maximum power density of the FISC is about 7000 W kg⁻¹ at an energy density of 33.90 Wh kg⁻¹. Such the energy storage performance is comparable or superior to that of most previously reported MXene based SCs (Fig. 6c) [64–70], and Mn&Fe based SCs (Table S1). As shown in Fig. S8, the thickness of FISCs is about 200 μm, so the total volume of the devices is about 0.075 cm³. Based on this, the maximum volumetric specific capacity of FISCs is calculated to be 185.3 mF cm⁻³, which can deliver a volumetric energy density up to 3.15 mWh cm⁻³. The FISCs can easily light up the LED with red color (Supplementary Video 1), confirming their promising potential as energy storage devices. Besides, the cycling stabilities of fabricated FISCs in the straight and bending states reveal the high retention above 80% after 10 000 cycles (Fig. 6e). Furthermore, an attempt to store renewable solar energy in FISCs has been made, in which the sustainable energy can be harvested for charging FISCs. As shown in Fig. 6f, after charging, the FISCs with high energy density could effectively supply power for potential practical applications. The corresponding video is shown in Supplementary Video 2. Therefore, this study introduces a new avenue for flexible energy storage devices, which is of great significance for the development of sustainable wearable and portable electronics.

Supplementary video related to this article can be found at <https://doi.org/10.1016/j.electacta.2019.135546>

4. Conclusions

In summary, we have successfully developed a new strategy to obtain a unique $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ electrode, which is realized by the confinement of MFNDs in partially oxidized $\text{Ti}_3\text{C}_2\text{T}_x$ to form a 3D interpenetrating assembly. Remarkably, oxidized $\text{Ti}_3\text{C}_2\text{T}_x$ with abundant active sites provides multiple and continuous conductive paths for the efficient charge transport, but also improves the structural stability of MFNDs in the electrode. As a result, the $\text{Ti}_3\text{C}_2\text{T}_x\text{@MFNDs}$ electrode exhibits superior electrochemical kinetics and energy storage behaviors in the EMIMBF₄ IL electrolyte. As a proof of concept, a kind of high-performance flexible ionogel SCs (FISCs) have been fabricated, presenting a high energy density (62.95 Wh kg⁻¹), a high power density (7000 W kg⁻¹), remarkable rate capability and long-term durability. More importantly, the as-fabricated devices can be also charged by harvesting sustainable energy and effectively supply power for practical applications. Therefore, this work provides a new insight into developing superior bimetallic oxide-based electrode materials for high-performance energy-storage devices.

Author contributions section

Minjie Shi: Conceptualization, Methodology, Writing-Original draft preparation. Mugilan Narayanasamy: Investigation, Writing-Original draft preparation. Cheng Yang: Writing- Reviewing and Editing. Jintian Jiang: Conceptualization. Liping Zhao: Resources. Subramania Angaiah: Resources. Chao Yan: Resources, Funding acquisition, Supervision.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.electacta.2019.135546>.

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